

Sitara Project B'd (Rev.A)

[1] Index

Rev	Date	Designer	Description
A	25.12.20	Ganghyeok Lim	Create design project

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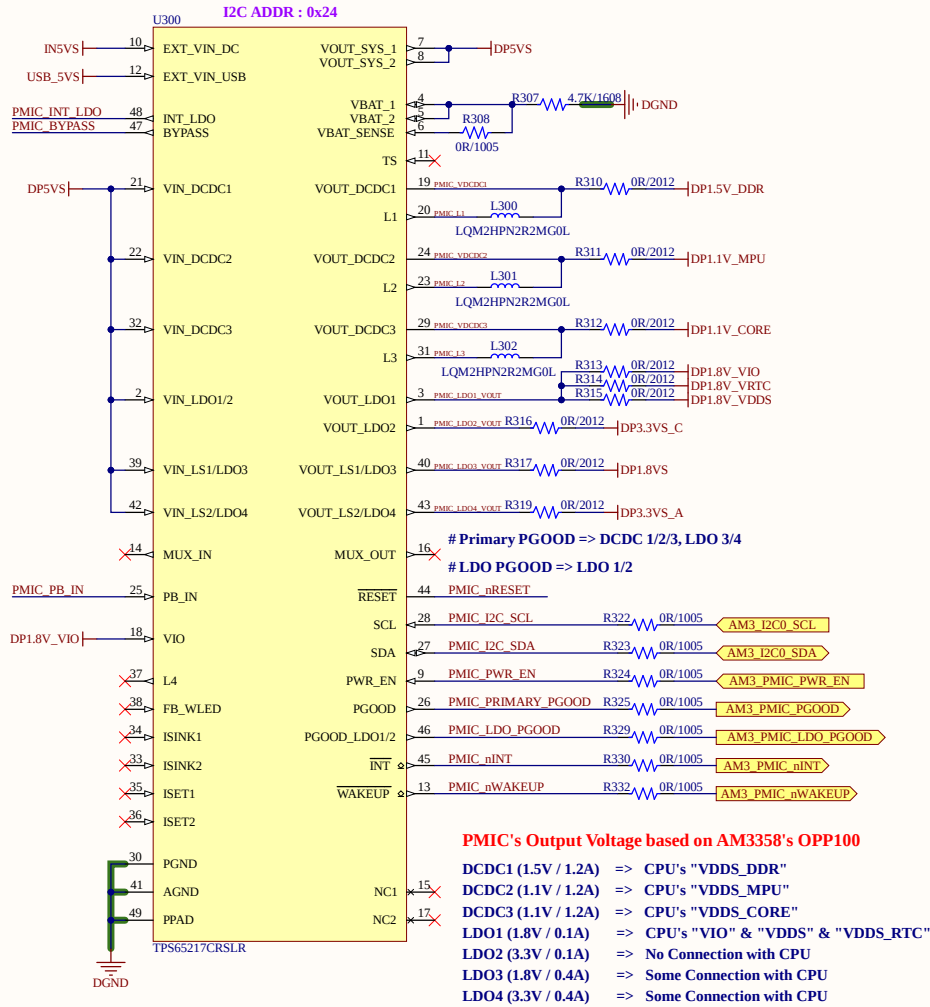
#8 Sitara I/O Bank 4/5
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Date		2025-12-28			
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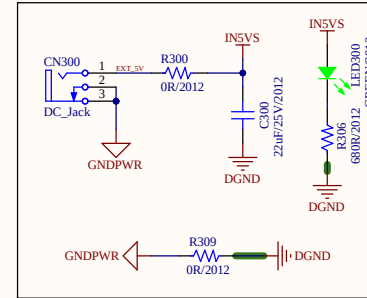
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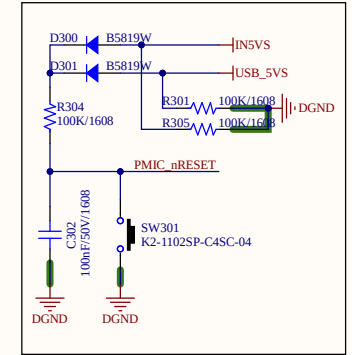
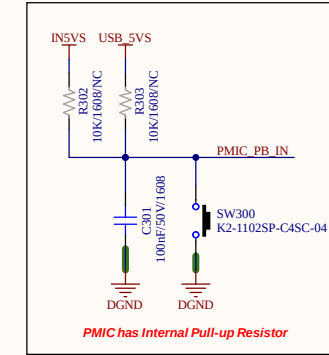
[3] Power



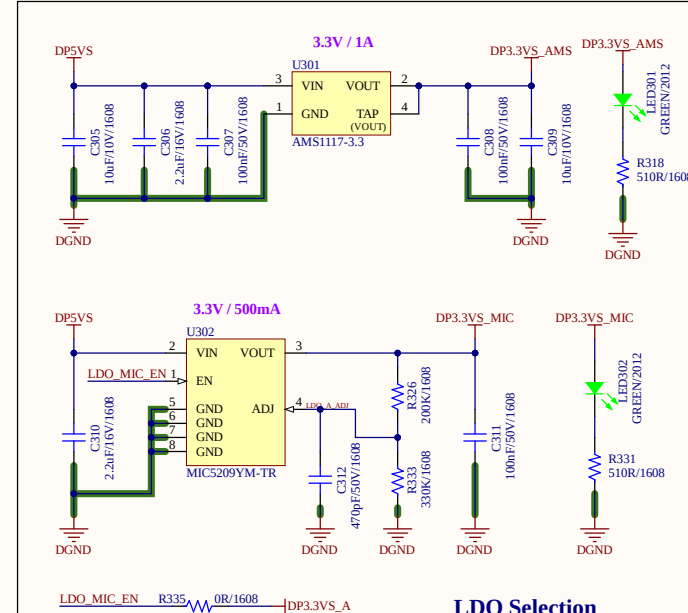
Power Connector (5V)



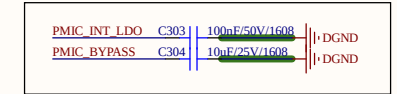
PMIC Power-On Button PMIC Reset Button



LDO (3.3V / 500mA)



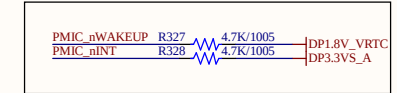
INT LDO / BYPASS Pin



I2C Pull-up Resistor



nWAKE-UP / nINT Pull-up

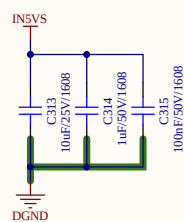


PMIC Power LED

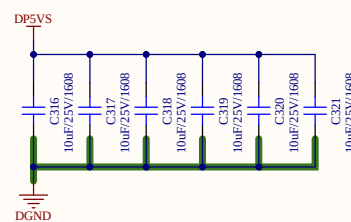


Decoupling Capacitors

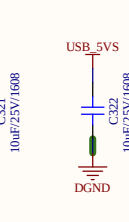
Decap for IN5VS



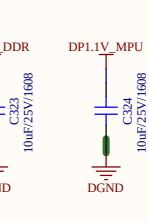
Decap for DP5VS



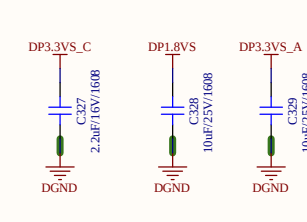
Decap for USB_5VS



Decap for DCDC 1/2/3

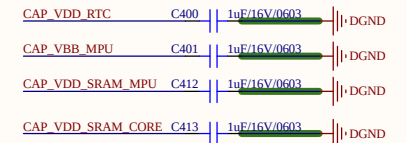
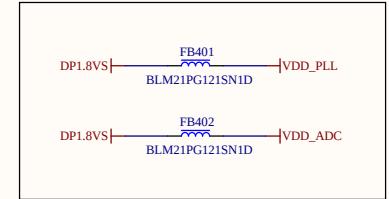
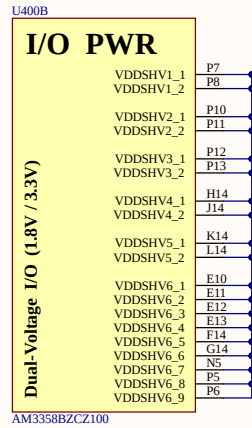


Decap for LDO 1~4



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Place Decaps Closely to Each Pin of PMIC



DP1.1V_CORE

DP1.1V_CORE

DGND

C448 100nF/25V/1608

C449 100nF/25V/1608

C450 100nF/25V/0603

C451 100nF/25V/0603

C452 100nF/25V/0603

C453 100nF/25V/0603

C454 100nF/25V/0603

C455 100nF/25V/0603

C456 100nF/25V/0603

C457 100nF/25V/0603

C458 100nF/25V/0603

C459 100nF/25V/0603

C460 100nF/25V/0603

C461 100nF/25V/0603

C462 100nF/25V/0603

C463 100nF/25V/0603

DP1.5V_DDR

DP1.5V_DDR

DGND

C464 100nF/25V/1608

C465 100nF/25V/1608

C466 100nF/25V/0603

C467 100nF/25V/0603

C468 100nF/25V/0603

C469 100nF/25V/0603

C470 100nF/25V/0603

C471 100nF/25V/0603

C472 100nF/25V/0603

DP1.8VS => VDDA1P8V_USB0/1, VDDS_SRAM_CORE_BG, VDDS_SRAM_MPU_BB (100nF)

VDD_PLL => VDDS_OSC(1uF), VDDS_PLL_MPU, VDDS_PLL_DDR (100nF)

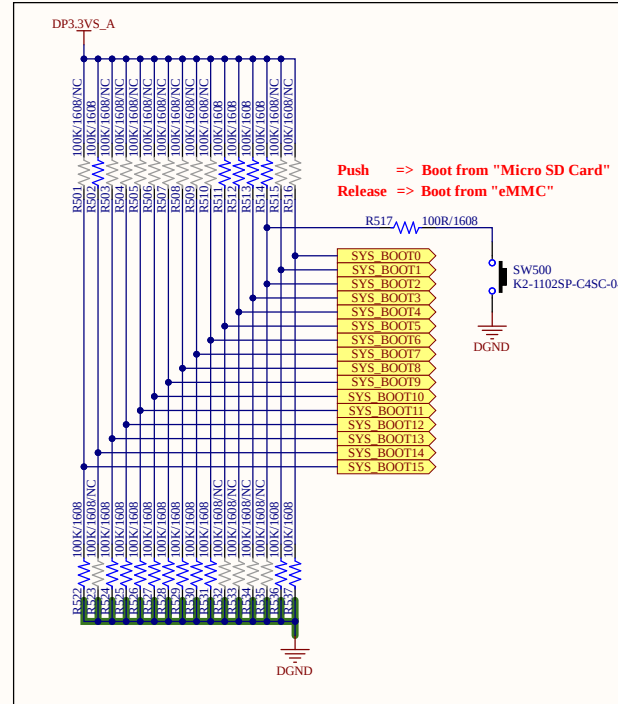
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Boot Config

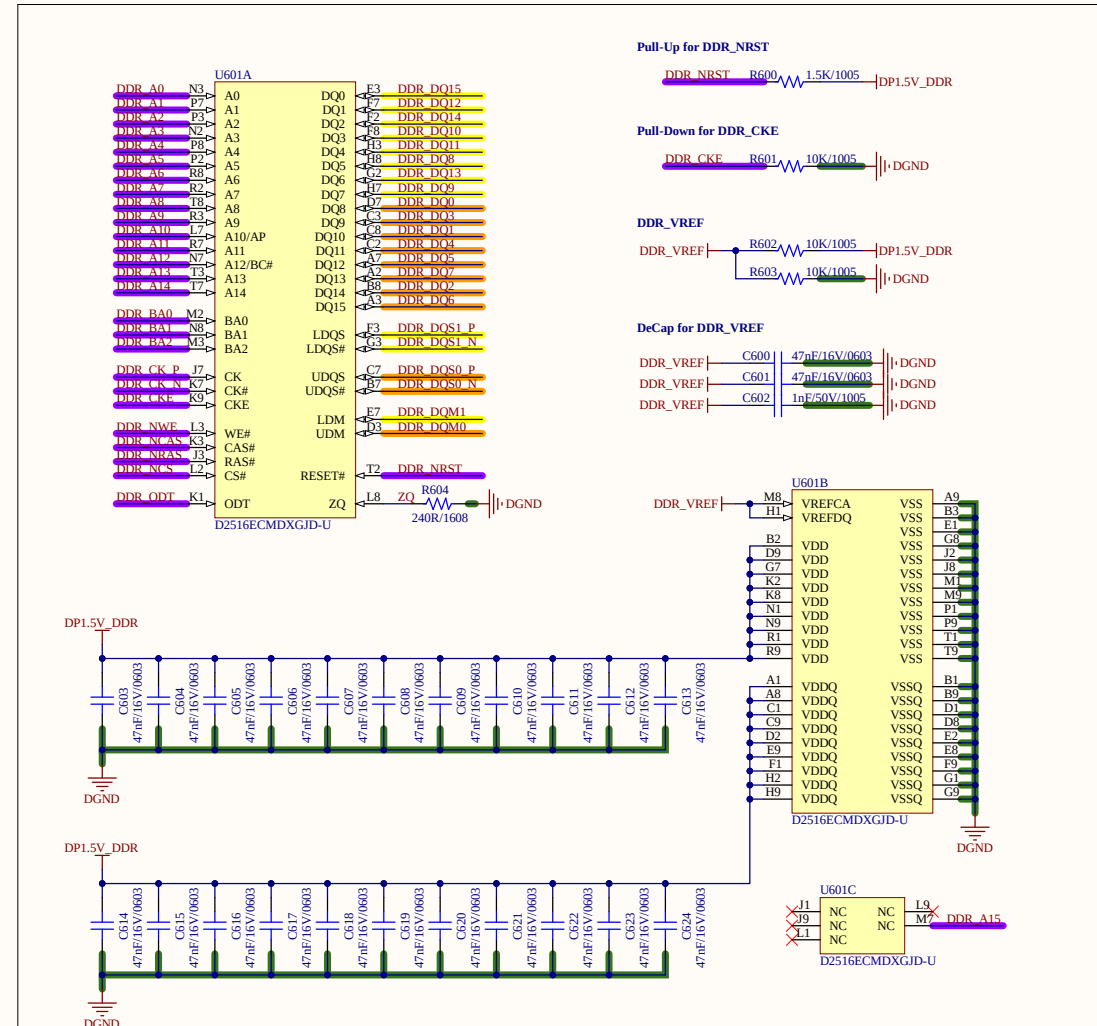


The diagram illustrates the USB0_VBUS connection for the UX60SC-MB-5ST(80) module. It shows the connection between the module's USB2.0 Mini-B connector (CN500) and the TPD4S012DRYR IC (U501). The module's VBUS pin is connected to the USB2.0 Mini-B connector's VBUS pin. The module's D+ pin is connected to the USB2.0 Mini-B connector's D+ pin. The module's D- pin is connected to the USB2.0 Mini-B connector's D- pin. The module's ID pin is connected to the USB2.0 Mini-B connector's ID pin. The module's GND pin is connected to the USB2.0 Mini-B connector's GND pin. The module's SHIELD pin is connected to the USB2.0 Mini-B connector's SHIELD pin. The module's UX60SC-MB-5ST(80) pin is connected to the USB2.0 Mini-B connector's UX60SC-MB-5ST(80) pin. The module's CGND pin is connected to the USB2.0 Mini-B connector's CGND pin. The module's USB0_VBUS pin is connected to the TPD4S012DRYR IC's VBUS pin. The module's USB0_D+ pin is connected to the TPD4S012DRYR IC's D+ pin. The module's USB0_D- pin is connected to the TPD4S012DRYR IC's D- pin. The module's USB0_ID pin is connected to the TPD4S012DRYR IC's ID pin. The module's USB0_GND pin is connected to the TPD4S012DRYR IC's GND pin. The module's USB0_SHIELD pin is connected to the TPD4S012DRYR IC's SHIELD pin. The module's USB0_UX60SC-MB-5ST(80) pin is connected to the TPD4S012DRYR IC's UX60SC-MB-5ST(80) pin. The module's USB0_CGND pin is connected to the TPD4S012DRYR IC's CGND pin.

RTC_KALDO_nEN R538 10K/1005 || DGND

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DDR3L Memory



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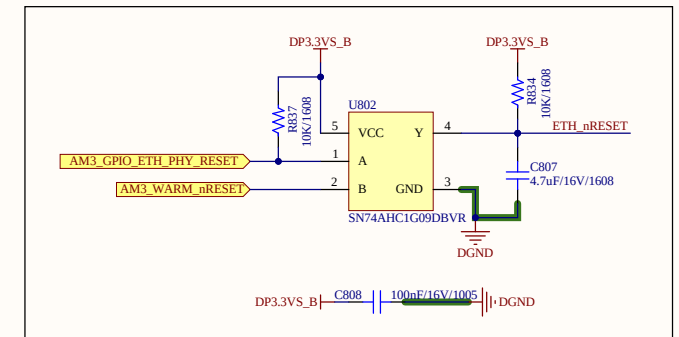
The schematic diagram illustrates the electrical connections for the MTF32GAZAHQD-WT board, featuring two main components: U703A and U702.

U703A (MTF32GAZAHQD-WT) Connections:

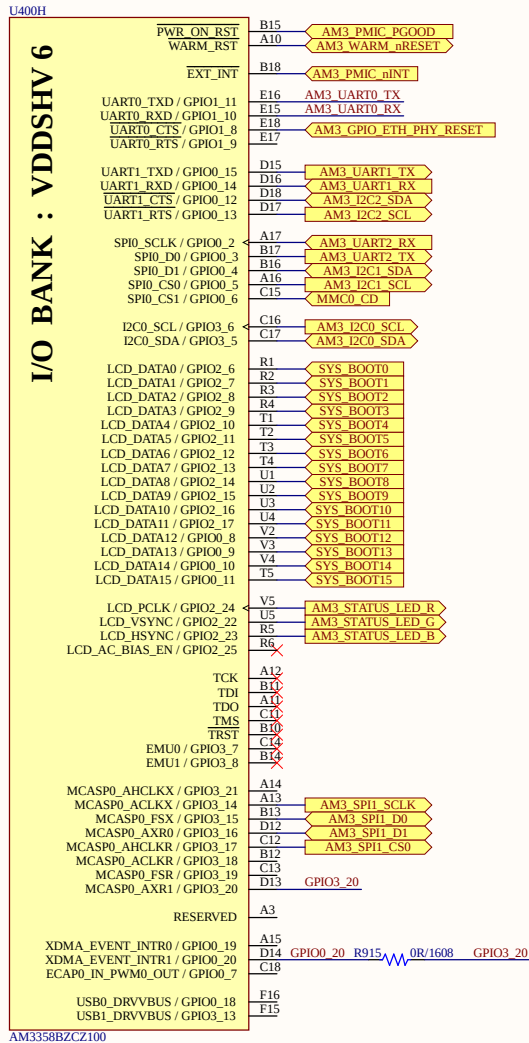
- Power and Ground:** VCC1, VCC2, VCC3, VCC4 are connected to DP1.8VS_eMMC. VCCQ1, VCCQ2, VCCQ3, VCCQ4, VCCQ5 are connected to DP1.8VS_eMMC. VSS1, VSS2, VSS3, VSS4, VSS5, VSS6, VSSQ1, VSSQ2, VSSQ3, VSSQ4, VSSQ5 are connected to DGND.
- Control Signals:** CLK, RST, CMD are connected to DP1.8VS_eMMC. DAT0 through DAT7 are connected to MMCI_0 through MMCI_7.
- Serial Resistor Placement:** A callout indicates "Place Serial Resistors Close to MPU". Resistors R717, R718, R719, R720, R721, R722, R723, R724, R725, R726, R727, R728, R729, R730, R731, R732, R733, R734, R735, R736, R737, R738, R739, R740, R741, R742, R743, R744, R745, R746, R747, R748, R749, R750, R751, R752, R753, R754, R755, R756, R757, R758, R759, R760, R761, R762, R763, R764, R765, R766, R767, R768, R769, R770, R771, R772, R773, R774, R775, R776, R777, R778, R779, R780, R781, R782, R783, R784, R785, R786, R787, R788, R789, R790, R791, R792, R793, R794, R795, R796, R797, R798, R799, R800, R801, R802, R803, R804, R805, R806, R807, R808, R809, R810, R811, R812, R813, R814, R815, R816, R817, R818, R819, R820, R821, R822, R823, R824, R825, R826, R827, R828, R829, R830, R831, R832, R833, R834, R835, R836, R837, R838, R839, R840, R841, R842, R843, R844, R845, R846, R847, R848, R849, R850, R851, R852, R853, R854, R855, R856, R857, R858, R859, R860, R861, R862, R863, R864, R865, R866, R867, R868, R869, R870, R871, R872, R873, R874, R875, R876, R877, R878, R879, R880, R881, R882, R883, R884, R885, R886, R887, R888, R889, R890, R891, R892, R893, R894, R895, R896, R897, R898, R899, R900, R901, R902, R903, R904, R905, R906, R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920, R921, R922, R923, R924, R925, R926, R927, R928, R929, R930, R931, R932, R933, R934, R935, R936, R937, R938, R939, R940, R941, R942, R943, R944, R945, R946, R947, R948, R949, R950, R951, R952, R953, R954, R955, R956, R957, R958, R959, R960, R961, R962, R963, R964, R965, R966, R967, R968, R969, R970, R971, R972, R973, R974, R975, R976, R977, R978, R979, R980, R981, R982, R983, R984, R985, R986, R987, R988, R989, R990, R991, R992, R993, R994, R995, R996, R997, R998, R999, R1000, R1001, R1002, R1003, R1004, R1005, R1006, R1007, R1008, R1009, R1010, R1011, R1012, R1013, R1014, R1015, R1016, R1017, R1018, R1019, R1020, R1021, R1022, R1023, R1024, R1025, R1026, R1027, R1028, R1029, R1030, R1031, R1032, R1033, R1034, R1035, R1036, R1037, R1038, R1039, R1040, R1041, R1042, R1043, R1044, R1045, R1046, R1047, R1048, R1049, R1050, R1051, R1052, R1053, R1054, R1055, R1056, R1057, R1058, R1059, R1060, R1061, R1062, R1063, R1064, R1065, R1066, R1067, R1068, R1069, R1070, R1071, R1072, R1073, R1074, R1075, R1076, R1077, R1078, R1079, R1080, R1081, R1082, R1083, R1084, R1085, R1086, R1087, R1088, R1089, R1090, R1091, R1092, R1093, R1094, R1095, R1096, R1097, R1098, R1099, R1100, R1101, R1102, R1103, R1104, R1105, R1106, R1107, R1108, R1109, R1110, R1111, R1112, R1113, R1114, R1115, R1116, R1117, R1118, R1119, R1120, R1121, R1122, R1123, R1124, R1125, R1126, R1127, R1128, R1129, R1130, R1131, R1132, R1133, R1134, R1135, R1136, R1137, R1138, R1139, R1140, R1141, R1142, R1143, R1144, R1145, R1146, R1147, R1148, R1149, R1150, R1151, R1152, R1153, R1154, R1155, R1156, R1157, R1158, R1159, R1160, R1161, R1162, R1163, R1164, R1165, R1166, R1167, R1168, R1169, R1170, R1171, R1172, R1173, R1174, R1175, R1176, R1177, R1178, R1179, R1180, R1181, R1182, R1183, R1184, R1185, R1186, R1187, R1188, R1189, R1190, R1191, R1192, R1193, R1194, R1195, R1196, R1197, R1198, R1199, R1200, R1201, R1202, R1203, R1204, R1205, R1206, R1207, R1208, R1209, R1210, R1211, R1212, R1213, R1214, R1215, R1216, R1217, R1218, R1219, R1220, R1221, R1222, R1223, R1224, R1225, R1226, R1227, R1228, R1229, R1230, R1231, R1232, R1233, R1234, R1235, R1236, R1237, R1238, R1239, R1240, R1241, R1242, R1243, R1244, R1245, R1246, R1247, R1248, R1249, R1250, R1251, R1252, R1253, R1254, R1255, R1256, R1257, R1258, R1259, R1260, R1261, R1262, R1263, R1264, R1265, R1266, R1267, R1268, R1269, R1270, R1271, R1272, R1273, R1274, R1275, R1276, R1277, R1278, R1279, R1280, R1281, R1282, R1283, R1284, R1285, R1286, R1287, R1288, R1289, R1290, R1291, R1292, R1293, R1294, R1295, R1296, R1297, R1298, R1299, R1300, R1301, R1302, R1303, R1304, R1305, R1306, R1307, R1308, R1309, R1310, R1311, R1312, R1313, R1314, R1315, R1316, R1317, R1318, R1319, R1320, R1321, R1322, R1323, R1324, R1325, R1326, R1327, R1328, R1329, R1330, R1331, R1332, R1333, R1334, R1335, R1336, R1337, R1338, R1339, R1340, R1341, R1342, R1343, R1344, R1345, R1346, R1347, R1348, R1349, R1350, R1351, R1352, R1353, R1354, R1355, R1356, R1357, R1358, R1359, R1360, R1361, R1362, R1363, R1364, R1365, R1366, R1367, R1368, R1369, R1370, R1371, R1372, R1373, R1374, R1375, R1376, R1377, R1378, R1379, R1380, R1381, R1382, R1383, R1384, R1385, R1386, R1387, R1388, R

A1	NC1	NG62	I2
A2	NC2	NG63	I3
A3	NC3	NG64	I4
A4	NC4	NG65	I5
A5	NC5	NG66	I6
A6	NC6	NG67	I7
A7	NC7	NG68	I8
A8	NC8	NG69	I9
A9	NC9	NC70	K1
B1	NC10	NC71	K2
B2	NC11	NC72	K3
B3	NC12	NC73	K4
B4	NC13	NC74	K5
B5	NC14	NC75	K6
B6	NC15	NC76	K7
B7	NC16	NC77	K8
B8	NC17	NC78	K9
B9	NC18	NC79	M1
B10	NC19	NC80	M2
B11	NC20	NC81	M3
B12	NC21	NC82	M4
B13	NC22	NC83	M5
B14	NC23	NC84	M6
B15	NC24	NC85	M7
B16	NC25	NC86	M8
B17	NC26	NC87	M9
B18	NC27	NC88	M10
B19	NC28	NC89	M11
B20	NC29	NC90	M12
B21	NC30	NC91	N1
B22	NC31	NC92	N2
B23	NC32	NC93	N3
B24	NC33	NC94	N4
B25	NC34	NC95	N5
B26	NC35	NC96	N6
B27	NC36	NC97	N7
B28	NC37	NC98	N8
B29	NC38	NC99	N9
B30	NC39	NC100	N10
B31	NC40	NC101	N11
B32	NC41	NC102	P1
B33	NC42	NC103	P2
B34	NC43	NC104	P3
B35	NC44	NC105	P4
B36	NC45	NC106	P5
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B38	NC47	NC108	P7
B39	NC48	NC109	P8
B40	NC49	NC110	P9
B41	NC50	VSF1	ER
B42	NC51	VSF2	ER
B43	NC52	VSF3	F1
B44	NC53	VSF4	F2
B45	NC54	VSF5	F3
B46	NC55	VSF6	F4
B47	NC56	VSF7	F5
B48	NC57	RFU1	A7
B49	NC58	RFU2	A8
B50	NC59	RFU3	A9
B51	NC60	RFU4	A10

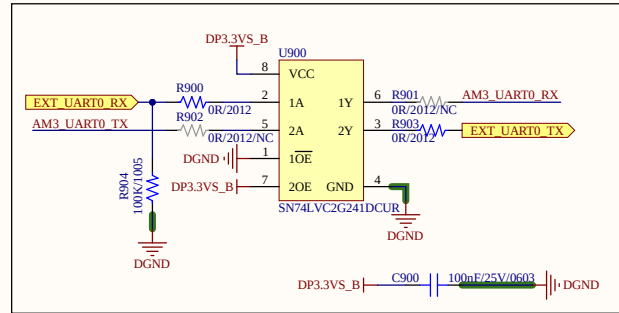
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[9] Sitara I/O Bank 6

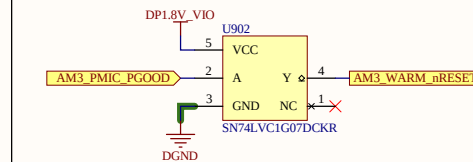


UART0 External Connector

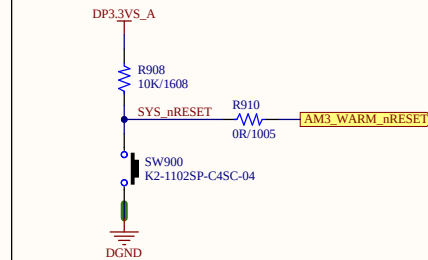


System Reset

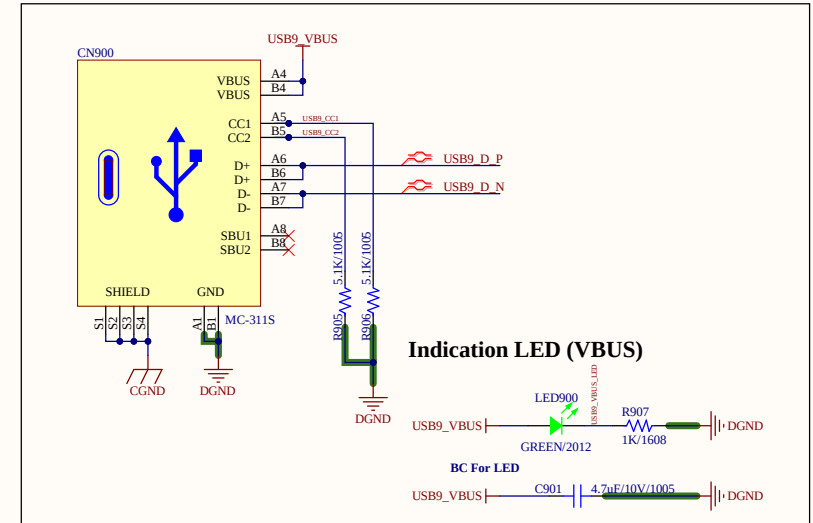
Power-On Reset



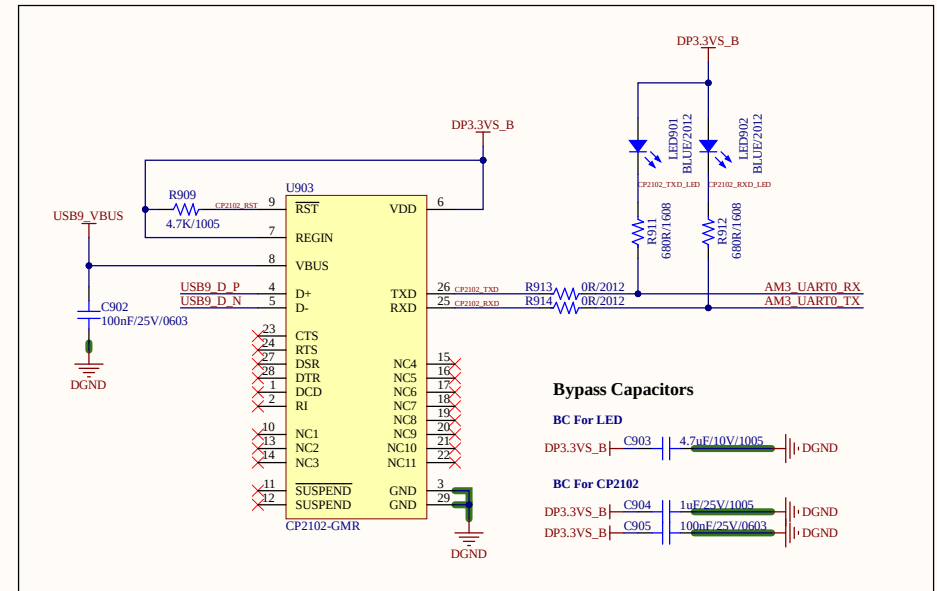
Reset Switch



USB Connector (Type-C)

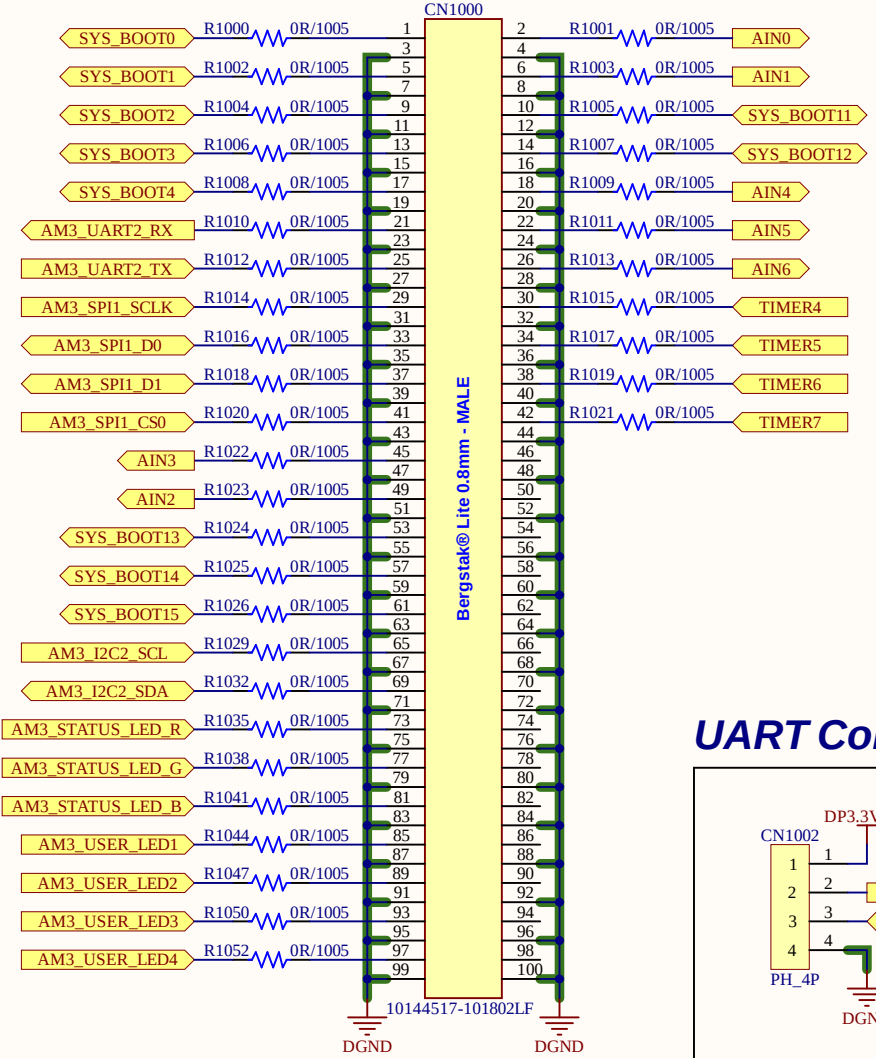
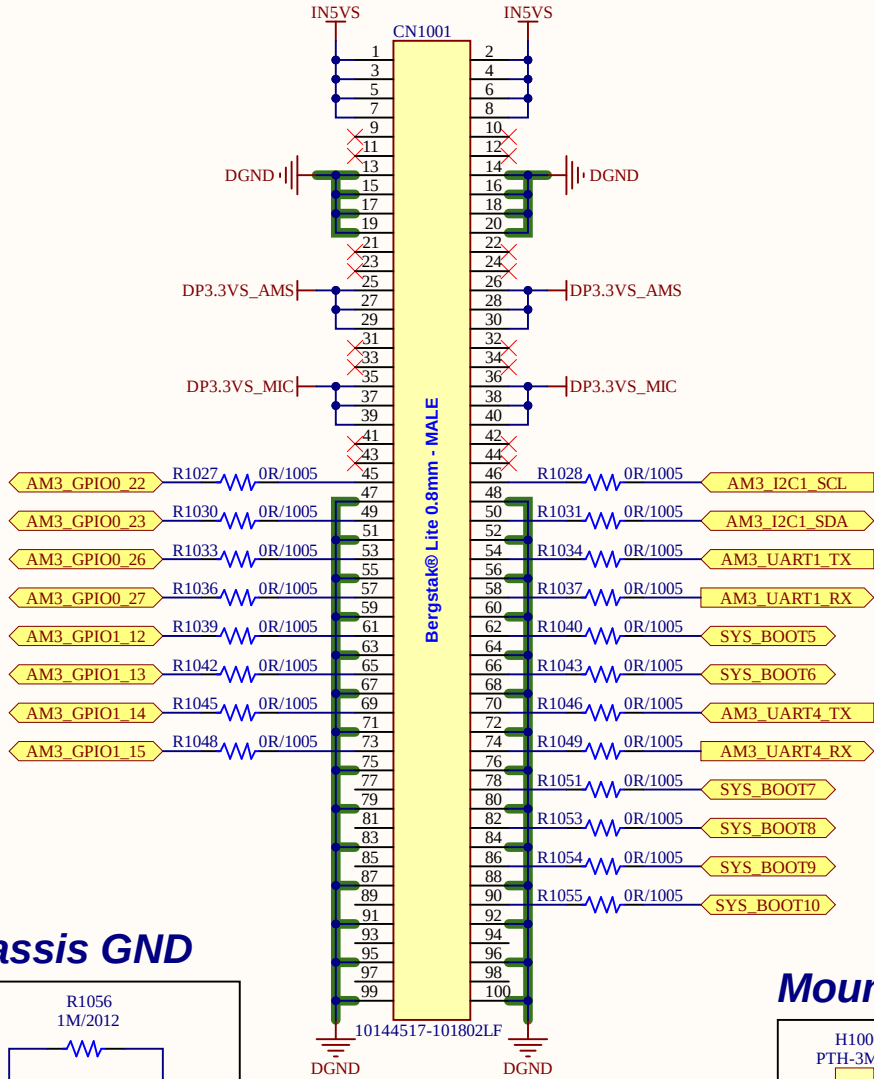


USB - UART Bridge

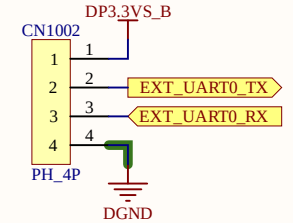


Title		Rev
Sitara_Project_Rev_A.PrjPcb		*
Doc	Sitara_IO_Bank_6.SchDoc	
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Date	2026-02-09	
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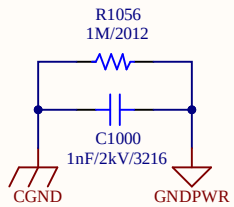
[10] Connectors



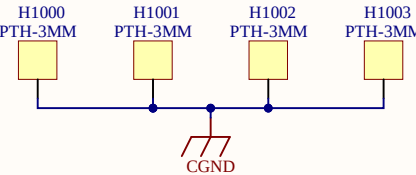
UART Connector



Chassis GND



Mount Holes



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